

Soldering Process and Safety Requirements

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Introduction

Soldering is a joining process commonly used in electronics to create both an electrical and mechanical connection between components. Unlike welding, the components themselves are not melted. Instead, a filler metal known as solder is heated until it becomes liquid and flows over the surfaces being joined. When the solder cools, it solidifies and creates a conductive connection. Correct soldering technique is important because poorly formed joints can cause increased electrical resistance, intermittent connections or complete circuit failure. Industry standards such as IPC J-STD-001 are used to define acceptable workmanship requirements for soldered electronic assemblies (NASA 2026).

Soldering Process

Before soldering begins, the work area and equipment should be prepared. Typical equipment includes a temperature-controlled soldering iron, solder wire, flux, soldering-iron stand, cleaning sponge or brass wool, component holders and appropriate personal protective equipment. Electronic components and printed circuit boards should be clean and free from grease, oxidation and contamination because contaminants may prevent the solder from properly bonding to the metal surfaces.

The soldering iron should then be brought to an appropriate operating temperature. For general electronic hand soldering, temperatures between approximately 325°C and 375°C are commonly used, although the correct temperature depends on the solder alloy, component and thermal characteristics of the circuit board (SparkFun Electronics 2026). Excessively high temperatures should be avoided because they can damage components and circuit-board tracks and increase the production of flux fumes.

Before creating a joint, the soldering-iron tip is tinned by applying a thin coating of molten solder. Tinning improves heat transfer between the iron and the joint and helps protect the tip from oxidation. The component should then be correctly positioned and secured so that it cannot move while the solder is cooling.

The soldering-iron tip is placed so that it simultaneously contacts the component lead and the copper pad. This allows both surfaces to reach the required temperature. Solder is then applied directly to the heated joint rather than simply being melted onto the iron. Once the correct temperature is reached, the solder melts and flows around the component lead and pad through a process known as wetting. Flux contained within the solder assists this process by removing surface oxides and improving solder flow.

Only enough solder to form a secure connection should be applied. The solder wire should be removed first, followed by the soldering iron. The connection should then remain stationary while it cools and solidifies. A correctly formed through-hole solder joint normally has a smooth, concave profile and demonstrates that the solder has adequately wetted both surfaces (SparkFun Electronics

2026). NASA soldering requirements similarly emphasise complete wetting and visual inspection of solder terminations as important workmanship criteria (NASA 2022).

Once soldering is complete, the joint should be visually inspected for defects such as insufficient solder, excessive solder, solder bridges, incomplete wetting or disturbed joints. Excess flux residue may also be removed using an appropriate electronics cleaning solution when required. Maintaining the soldering tip by cleaning and re-tinning it before the iron is switched off also helps prevent oxidation and extends tip life.

Safety Requirements

One of the most obvious soldering hazards is heat. The tip of a soldering iron can exceed 300°C and can cause significant burns or ignite combustible materials. The soldering iron should therefore always be returned to a stable, heat-resistant stand when it is not being used. The operator should never touch the heated section of the iron and should keep loose clothing, cables and other flammable materials away from the tip.

Eye protection is also recommended because molten solder or flux may occasionally spit or splash when heated. Components should preferably be held with a vice, clamp or suitable PCB holder rather than by hand while soldering.

Another significant hazard is exposure to soldering fumes. Rosin or colophony-based flux releases fumes when heated, and repeated occupational exposure has been associated with respiratory irritation and occupational asthma. The UK Health and Safety Executive identifies rosin-based solder flux fumes as an important cause of occupational asthma and recommends using effective fume extraction, avoiding overheating and keeping the operator's head away from the rising fume plume (HSE 2025).

For this reason, soldering should be undertaken in a well-ventilated area with local exhaust ventilation or an appropriate solder-fume extraction system positioned close to the joint. Simply working in a large room does not guarantee that fumes will remain outside the operator's breathing zone.

Additional precautions are required when lead-based solder is used. Lead may enter the body through inhalation or ingestion of contamination transferred from the hands to food or the mouth. NIOSH recommends substituting lead-containing materials with less hazardous alternatives where practical and using engineering controls, good housekeeping and suitable hygiene procedures where lead exposure remains possible (NIOSH 2024). Lead-free solder should therefore be considered where appropriate.

Operators should wash their hands thoroughly after soldering, particularly before eating or drinking, and food should not be consumed at the soldering workstation. Work surfaces should also be cleaned regularly to prevent the accumulation or transfer of contamination. Safety Data Sheets should be consulted for the specific solder, flux and cleaning chemicals being used.

Electronic components can also be damaged by electrostatic discharge (ESD). When working with ESD-sensitive devices, appropriate precautions such as an antistatic mat and grounded wrist strap should be used. NASA workmanship guidance recognises ESD control as an important part of the handling and assembly of electronic hardware (NASA 2026).

Conclusion

Successful soldering requires both correct workmanship and appropriate safety controls. A reliable solder joint is produced by preparing clean surfaces, using an appropriately heated and tinned soldering iron, heating the component and pad together, applying the correct amount of solder and inspecting the completed connection. At the same time, operators must control hazards associated with high temperatures, flux fumes, lead contamination and electrostatic discharge. Appropriate ventilation, good hygiene, correct equipment and PPE allow soldering to be performed safely while reducing the likelihood of component damage or unreliable electrical connections.

Bibliography

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